ME488 Homework 1

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Chapter 2 Problems

2.1 Basic descriptive statistics / standard error / variance

2.7 Normality + one-sample inference

2.10 Interpreting hypothesis-test output

2.15 Two-sample t-test output and interpretation

2.19 One-sample t-test in an engineering context

2.20 Normality assumption and robustness

2.21 Full one-sample t-test + interpretation

2.23 Two-sample comparison with known, unequal σ's

2.25 Two-sample inference / practical interpretation

2.29 Two-sample engineering experiment

Problem 2.1

Problem Statement:

Computer output for a random sample of data is shown below. Some of the quantities are missing. Compute the values of the missing quantities.

Variable	N	Mean	SE Mean	Std. Dev.	Variance	Minimum	Maximum
Y	9	19.96	NA	3.12	NA	15.94	27.16

Solution:

```
# Import relevant data from problem statement
nn <- c(9)
stdDevResult <- c(3.12)

# Calculate SE Mean and variance
SEM <- stdDevResult/sqrt(nn)
variance <- stdDevResult^2
```

Conclusion:

Variable	N	Mean	SE Mean	Std. Dev.	Variance	Minimum	Maximum
Y	9	19.96	1.04	3.12	9.7344	15.94	27.16

Problem 2.7

Problem Statement:

Consider the following sample data:

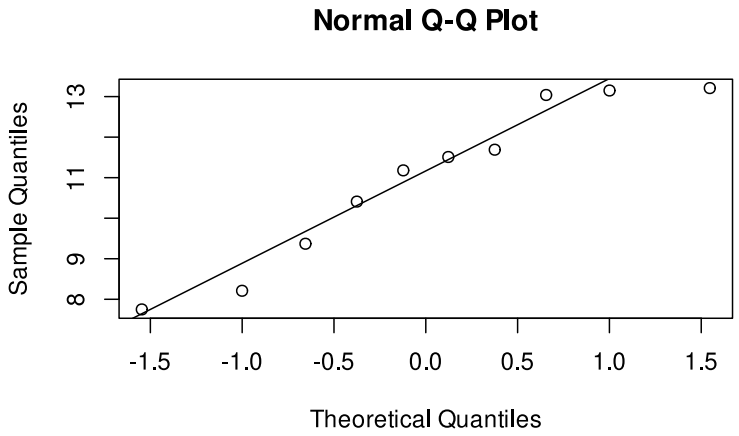
```
9.37, 13.04, 11.69, 8.21, 11.18, 10.41, 13.15, 11.51, 13.21, 7.75
```

Is it reasonable to assume that this data is a sample from a normal distribution? Is there evidence to support a claim that the mean of the population is 10?

Solution:

To test if the data is normally distributed, I will make a normal QQ plot to see if the data points lie in normal quantiles.

```
p7Data = c(9.37, 13.04, 11.69, 8.21, 11.18, 10.41, 13.15, 11.51, 13.21, 7.75)
qqnorm(p7Data)
qqline(p7Data)
```



Plotting the normal QQ plot suggests that the data set is normally distributed, although the 13.21 data point is a slight outlier. Continuing with the assumption that the data is normally distributed, I will run a one-sample two-sided t-test with a standard 95% confidence interval in order to test if the mean of 10 lies inside the normally distributed data.

```r
t.test(p7Data, mu = 10) # Defaults to two-sided 95% confidence test
```

```
    One Sample t-test

data:  p7Data
t = 1.5102, df = 9, p-value = 0.1653
alternative hypothesis: true mean is not equal to 10
95 percent confidence interval:
  9.525956 12.378044
sample estimates:
mean of x
   10.952
```

The t-test tells us that there is a 95% chance that the mean is between 9.52 and 12.37, which is statistically significant enough to reject the alternative hypothesis that the mean is not equal to 10.

Conclusion:

It is statistically likely that the data set is normally distributed and that the mean does not differ from 10.

Problem 2.10

Problem Statement:

A two-sample t-test has been conducted and the sample sizes are $n_1 = n_2 = 10$. The computed value of the test statistic is to $t_0 = 2.15$. If the null hypothesis is two-sided, an upper bound on the P-value is:

(a) 0.10 (b) 0.05 (c) 0.025 (d) 0.01 (e) None of the above

Solution:

First I want to find out what P-value is associated with t_0. Since each sample has $n - 1$ degrees of freedom, the combined degrees of freedom for the two sample t-test is 18. I can then use R's distribution function to determine the P-value of t_0 approached from the upper tail.

```
pt(2.15, df = 18, lower.tail = FALSE)
```

```
[1] 0.02269846
```

This function outputs the associated P-value as 0.0227. The nearest upper bound from the list of answers in the problem statement is 0.025. Testing this P-value with R's quantile function yields the following t-value:

```
qt(0.025, df = 18, lower.tail = FALSE)
```

```
[1] 2.100922
```

As expected, this t-value is lower than t_0 because $p = 0.025$ contains a slightly smaller portion of the distribution.

Conclusion:

`(c) 0.025` is the most reasonable upper bound for the P-value.

Problem 2.15

Problem Statement:

Two-Sample t-Test and Cl: Y1, Y2

```
Two-sample T for Y1 vs Y2
    n Mean Std. Dev. SE Mean
Y1 20 50.19 1.71 0.38
Y2 20 52.52 2.48 0.55
Difference = mu (X1) - mu (X2)
Estimate for difference: -2.33341
95% CI for difference: (-3.69547, -0.97135)
T-Test of difference = 0 (vs not =): T-Value = -3.47
P-Value = 0.001 DF = 38
Both use Pooled Std. Dev. = 2.1277
```

(a) Can the null hypothesis be rejected at the 0.05 level? Why?

(b) Is this a one-sided or a two-sided test?

(c) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$, would you reject the null hypothesis at the 0.05 level?

(d) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 < 2$, would you reject the null hypothesis at the 0.05 level? Can you answer this question without doing any additional calculations? Why?

(e) Use the output and the t table to find a 95 percent upper confidence bound on the difference in means.

(f) What is the P-value if the hypotheses are $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$?

Solution:

(a):

The P-value of 0.001 is less than the significance level of $\alpha = 0.05$, so I would reject the null hypothesis.

(b):

This is a two-sided t-test because it is testing if the alternative hypothesis is not equal to zero, meaning that it is testing both if the the hypothesis is greater than or less than 0.

(c):

Because the hypothesis has changed, I need to calculate a new test statistic to compare against. Equation 2.24 from our textbook (Douglas C. Montgomery - Design and Analysis of Experiments 10th Ed.) calculates the test statistic for the hypothesis $H_0 : \mu_1 - \mu_2 = 0$:

$$t_0 = \frac{(\bar{y}_1 - \bar{y}_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

This equation can be modified to test $H_0 : \mu_1 - \mu_2 = 2$ by offsetting the difference of the means:

$$t_0 = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Then we can use the new value for t_0 to calculate the two-sided P-value to either reject or fail the reject the new null hypothesis.

```
# Given information
difference <- -2.33341
pooled_sd <- 2.1277
n1 <- 20
n2 <- 20
delta <- 2

# Calculate new t-statistic
t0 <- (difference - delta) / (pooled_sd * sqrt(1/n1 + 1/n2))

# Calculate two-sided P-value
2 * pt(-abs(t0), df = 38)
```

```
[1] 1.418836e-07
```

The new P-value is significantly less than 0.05, so the I reject the null hypothesis.

(d):

The null hypothesis has not changed from the part (c), the it can once again be rejected using the same calculation from before. The new alternative hypothesis is left-handed because it states that the difference of the means is less than 2. This alternative hypothesis is supported by the original 95% confidence interval that the difference was between −3.69547 and −0.97135.

(e):

We can calculate an upperbound at a 95% confidence interval using equation 2.30 from the textbook:

$$\mu_1 - \mu_2 \leq \bar{y}_1 - \bar{y}_2 + t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

```r
# Given information
difference <- -2.33341
pooled_sd <- 2.1277
n1 <- 20
n2 <- 20

t_a <- qt(0.95, df=38)

upBound <- difference + t_a * (pooled_sd * sqrt(1/n1 + 1/n2))
upBound
```

```
[1] -1.199036
```

(f):

I will calculate the new two-sided P-Value using the t-value I calculated in part (c)

```r
# Given information
difference <- -2.33341
pooled_sd <- 2.1277
n1 <- 20
n2 <- 20
delta <- 2

t0 <- (difference - delta) / (pooled_sd * sqrt(1/n1 + 1/n2))
2 * pt(-abs(t0), 38)
```

```
[1] 1.418836e-07
```

Problem 2.19

Problem Statement:

The shelf life of a carbonated beverage is of interest. Ten bottles are randomly selected and tested, and the following results are obtained:

```
Shelf Life (days)
108, 124, 124, 106, 115, 138, 163, 159, 134, 139
```

(a) We would like to demonstrate that the mean shelf life exceeds 120 days. Set up appropriate hypotheses for investigating this claim.

(b) Test these hypotheses using a = 0.01. What are your conclusions?

(c) Find the P-value for the test in part (b).

(d) Construct a 99 percent confidence interval on the mean shelf life.

Solution:

(a)

My hypotheses are $H_0 : \mu = 120$ and $H_1 : \mu > 120$

(b)

```r
p19Data <- c(108, 124, 124, 106, 115, 138, 163, 159, 134, 139)
t.test(p19Data, alternative = "greater", mu = 120, conf.level = .99)
```

```
    One Sample t-test

data:  p19Data
t = 1.7798, df = 9, p-value = 0.05441
alternative hypothesis: true mean is greater than 120
99 percent confidence interval:
 113.5618      Inf
sample estimates:
mean of x
      131
```

(c)

$p = 0.05441$

(d)

My last t-test used a one-sided t-test. In order to calculate a realistic range for possible means with the existing data set, I will switch to using a two-sided test to calculate the confidence interval.

```r
p19Data <- c(108, 124, 124, 106, 115, 138, 163, 159, 134, 139)
t.test(p19Data, mu = 120, conf.level = .99)
```

```
    One Sample t-test

data:  p19Data
t = 1.7798, df = 9, p-value = 0.1088
alternative hypothesis: true mean is not equal to 120
99 percent confidence interval:
 110.914 151.086
sample estimates:
mean of x
      131
```

The 99% confidence interval puts the mean somewhere between 110.91 and 151.09

Problem 2.20

Problem Statement:

Consider the shelf life data in Problem 2.19. Can shelf life be a described or modeled adequately by a normal distribution? What effect would the violation of this assumption have on the test procedure you used in solving Problem 2.15?

Solution:

Assuming ideal conditions like a consistent climate controlled environment, it makes sense to model shelf life as a normal distribution because most food spoils around the same time without trending towards shorter or longer times.

Violating this assumption would mean that my previous solution process is invalid since the t-test used was meant for a normal distribution, also invalidating further analysis based on that test.

Problem 2.21

Problem Statement:

The time to repair an electronic instrument is a normally distributed variable measured in hours. The repair times for 16 such instruments chosen at random are as follows:

(a) You wish to know if the mean repair time exceeds 225 hours. Set up appropriate hypotheses for investigating this issue.

(b) Test the hypotheses you formulated in part (a). What are your conclusions? Use $\alpha = 0.05$.

(c) Find the P-value for the test.

(d) Construct a 95% confidence interval on mean repair time.

```
Hours
159, 224, 222, 149, 280, 379, 362, 260, 101, 179, 168, 485, 212, 264, 250, 170
```

Solution:

(a)

My hypotheses are $H_0 : \mu = 225$ and $H_1 : \mu > 225$

(b)

```
p21Data <- c(159, 224, 222, 149, 280, 379, 362, 260, 101, 179, 168, 485, 212,
264, 250, 170)
t.test(p21Data, alternative = "greater", mu = 225)
```

```

    One Sample t-test

data:  p21Data
t = 0.66852, df = 15, p-value = 0.257
alternative hypothesis: true mean is greater than 225
95 percent confidence interval:
 198.2321      Inf
sample estimates:
mean of x
    241.5
```

(c)

$p = 0.257$

(d)

```
p21Data <- c(159, 224, 222, 149, 280, 379, 362, 260, 101, 179, 168, 485, 212,
264, 250, 170)
t.test(p21Data, mu = 225)
```

```

    One Sample t-test

data:  p21Data
t = 0.66852, df = 15, p-value = 0.514
alternative hypothesis: true mean is not equal to 225
95 percent confidence interval:
 188.8927 294.1073
sample estimates:
mean of x
    241.5
```

The 95% confidence interval puts the mean somewhere between 188.89 and 294.11

Problem 2.23

Problem Statement:

Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The filling processes can be assumed to be normal, with standard deviations of $\sigma_1 = 0.015$ and $\sigma_2 = 0.018$. The quality engineering department suspects that both machines fill to the same net volume, whether or not this volume is 16.0 ounces. An experiment is performed by taking a random sample form the output of each machine.

(a) State the hypotheses that should be tested in this experiment.

(b) Test these hypotheses using $\alpha = 0.05$. What are your conclusions?

(c) Find the P-Value for this test.

(d) Find a 95% confidence interval on the difference in mean fill volume for the two machines.

Machine 1	Machine 2
16.03	16.02
16.04	15.97
16.05	15.96
16.05	16.01
16.02	15.99
16.01	16.03
15.96	16.04
15.98	16.02
16.02	16.01
15.99	16.00

Solution:

(a)

$H_0 : \mu_1 - \mu_2 = 0$ versus $H_1 : \mu_1 - \mu_2 \neq 0$,

(b)

```
# Import libraries and problem data
library(BSDA)
library(readxl)
p23Data <- read_excel("../homework_data.xlsx", sheet = "Chapter 02", range =
"E3:F13")

z.test(
  p23Data[1],
  p23Data[2],
  sigma.x = 0.015,
  sigma.y = 0.018,
  mu = 0,
  alternative = "two.sided",
  conf.level = 0.95
)
```

```
    Two-sample z-Test

data:  p23Data[1] and p23Data[2]
z = 1.3496, p-value = 0.1771
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.004522262  0.024522262
sample estimates:
mean of x mean of y
   16.015    16.005
```

(c)

$p = 0.1771$

(d)

The 95% confidence interval puts the mean somewhere between −0.00452 and 0.0245

Problem 2.25

Problem Statement:

The following are the burning times (in minutes) of chemical flares of two different formulations. The design engineer are interested in both the mean and variance of the burning times.

(a) Test the hypothesis that the two variances are equal. Use $\alpha = 0.05$.

(b) Using the results of (a), test the hypothesis that the mean burning times are equal. Use $\alpha = 0.05$. What is the P-value for this test?

(c) Discuss the role of the normality assumption in this problem. Check the assumption of normality for both types of flares.

Type 1	Type 2
65	64
81	71
57	83
66	59
82	65
82	56
67	69
59	74
75	82
70	79

Solution:

(a)

```
library(readxl)
p25Data <- read_excel("../homework_data.xlsx", sheet = "Chapter 02", range =
"H3:I13")

var.test(p25Data[[1]], p25Data[[2]])
```

```
    F test to compare two variances

data:  p25Data[[1]] and p25Data[[2]]
F = 0.97822, num df = 9, denom df = 9, p-value = 0.9744
```

```
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2429752 3.9382952
sample estimates:
ratio of variances
         0.9782168
```

$p = 0.9744$ which is significantly greater than $\alpha = 0.05$, so we fail to reject the null hypothesis of $H_0 : \sigma_1^2 = \sigma_2^2$, which suggests that the two variances may be equal.

(b)

Because we can't reject the null hypothesis, it is fair to continue with the assumption that the variances are equal. This information can be used in a subsequent t-test to test if there is difference between the means.

```
library(readxl)
p25Data <- read_excel("../homework_data.xlsx", sheet = "Chapter 02", range =
"H3:I13")

t.test(p25Data[[1]], p25Data[[2]], var.equal = TRUE)
```

```

    Two Sample t-test

data:  p25Data[[1]] and p25Data[[2]]
t = 0.048008, df = 18, p-value = 0.9622
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -8.552441  8.952441
sample estimates:
mean of x mean of y
     70.4      70.2
```

$p = 0.9622$, which is significantly greater than 0.05, so we fail to reject the null hypothesis that the means differ.

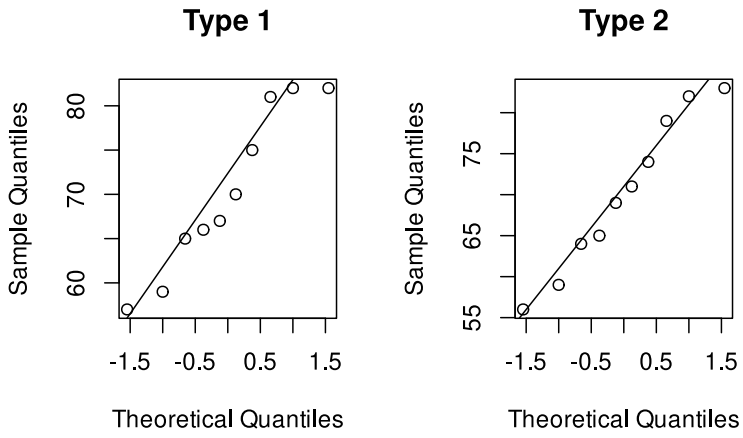
(c)

```
library(readxl)
p25Data <- read_excel("../homework_data.xlsx", sheet = "Chapter 02", range =
"H3:I13")

par(mfrow = c(1, 2))
```

```r
qqnorm(p25Data[[1]], main = "Type 1")
qqline(p25Data[[1]])

qqnorm(p25Data[[2]], main = "Type 2")
qqline(p25Data[[2]])
```




```r
par(mfrow = c(1, 1))
```

Both the F and T test rely on normally distributed data to be reliable. Plotting the normal QQ plots for each flare (shown above), they each appear to be normally distributed, meaning this data can be trusted.

Problem 2.29

Problem Statement:

The diameter of a ball bearing was measured by 12 inspectors, each using two different kinds of calipers. The results are as follows:

(a) Is there a significant difference between the means of the population of measurements from which the two samples were selected? Use $\alpha = 0.05$.

(b) Find the P-value for the test in part (a).

(c) Construct a 95% confidence interval on the difference in mean diameter measurements for the two types of calipers.

Caliper 1	Caliper 2
0.265	0.264
0.265	0.265
0.266	0.264
0.267	0.266
0.267	0.267
0.265	0.268
0.267	0.264
0.267	0.265
0.265	0.265
0.268	0.267
0.268	0.268
0.265	0.269

Solution:

(a)

```
# Import problem data
library(readxl)
p29Data <- read_excel("../homework_data.xlsx", sheet = "Chapter 02", range =
"T3:U15")

t.test(p29Data[[1]], p29Data[[2]], paired = TRUE)
```

```
    Paired t-test

data:  p29Data[[1]] and p29Data[[2]]
t = 0.43179, df = 11, p-value = 0.6742
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.001024344  0.001524344
sample estimates:
mean difference
        0.00025
```

$p = 0.6742$, which is significantly greater than 0.05, so we fail to reject the null hypothesis that the means are different.

(b)

$p = 0.6742$

(c)

The 95% confidence interval puts the mean somewhere between −0.00102 and 0.00152